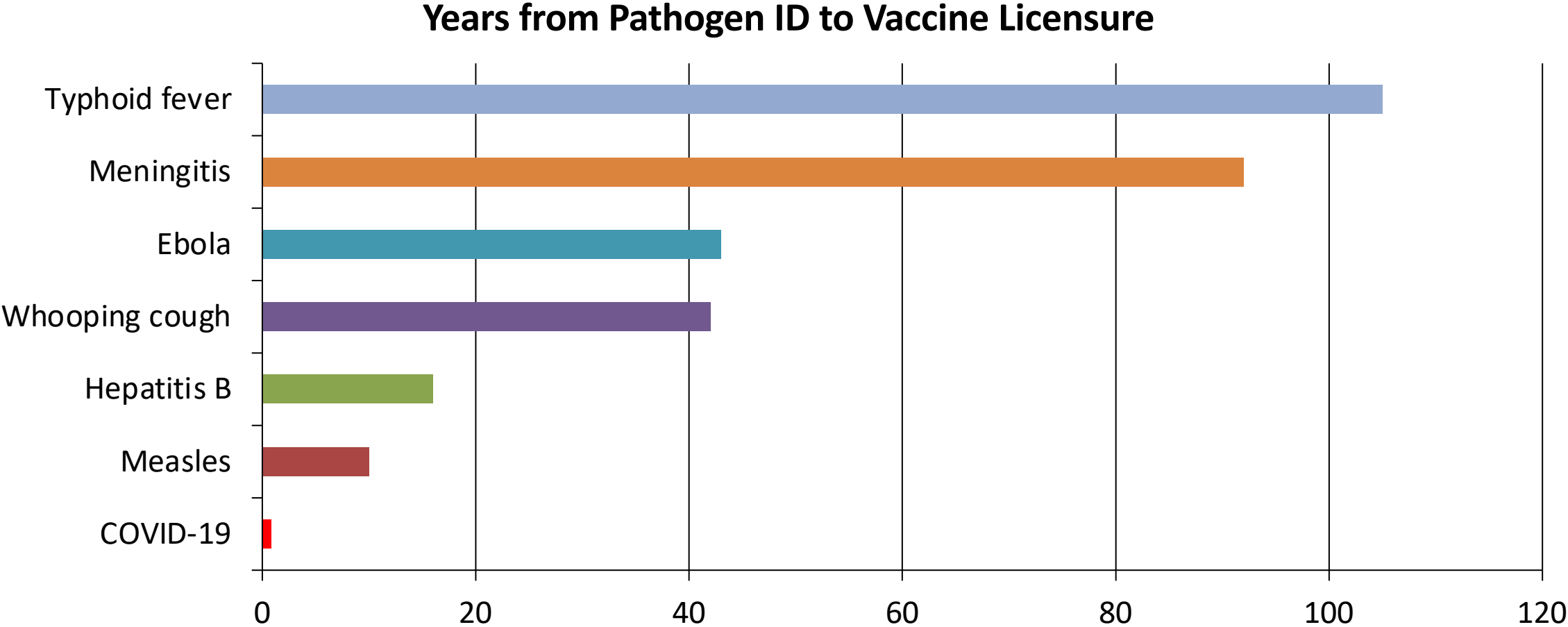


The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

A data-driven overview of public health
transformation (1796–2020)

Source: Our World in Data

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



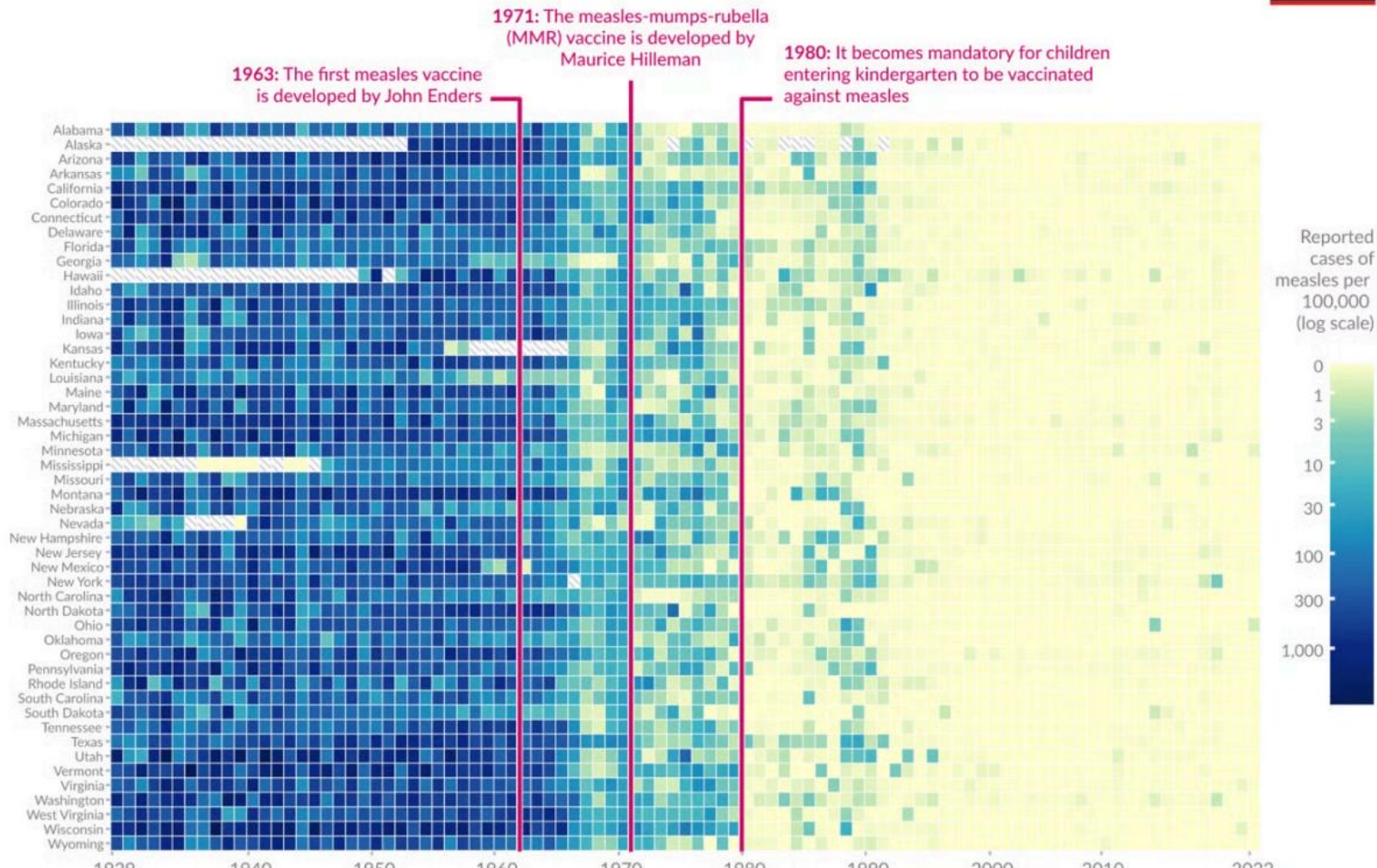
100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-Vaccine Cases (per mil/yr)	Post-Vaccine Cases (per mil/yr)	Case Reduction	Pre-Vaccine Deaths (per mil/yr)	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Measles	3,044	0.2	99.99%	2.5	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Vaccines reduced measles cases across US states

Our World
in Data



1963: First measles vaccine

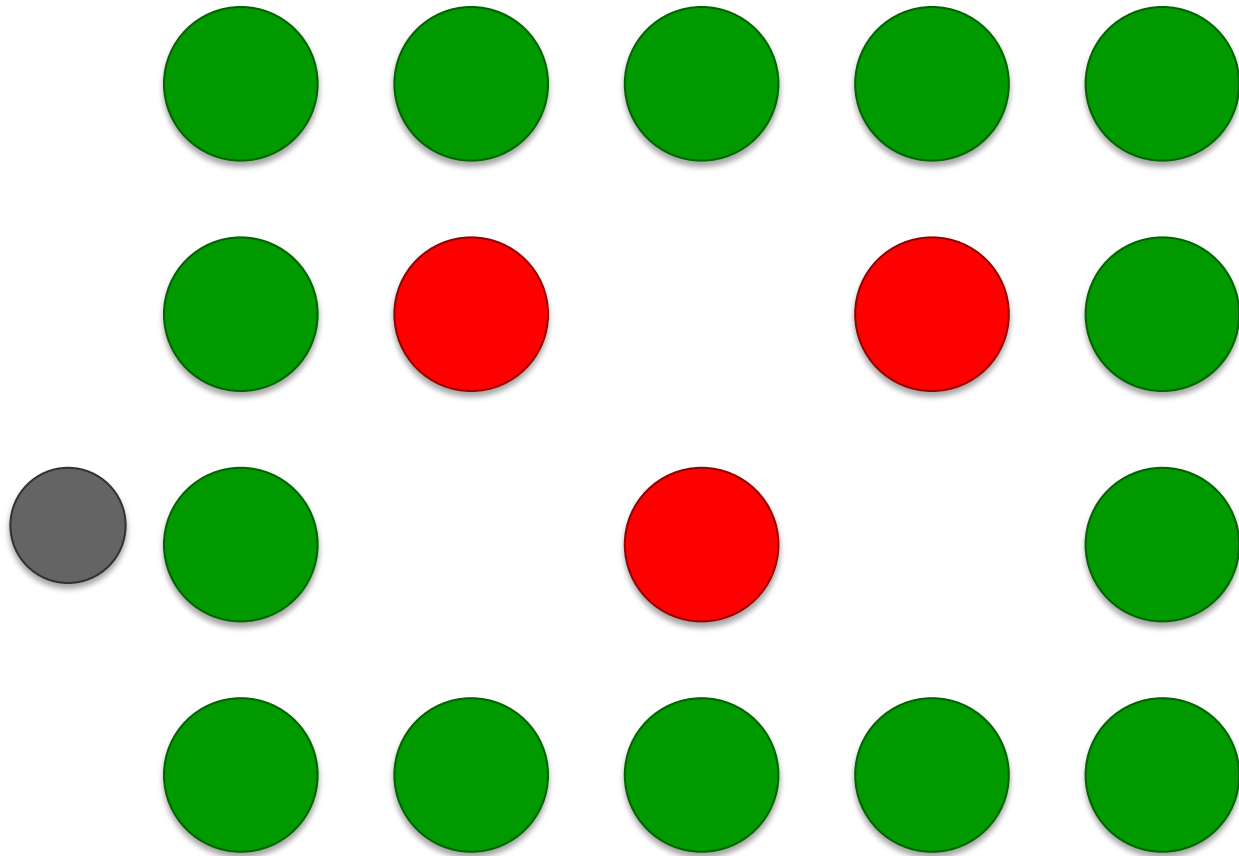
1971: MMR vaccine introduced

1980: Mandated for kindergarten

Smallpox: The Only Human Disease Eradicated Through Vaccination

- The Greatest Vaccination Success Story
 - 1959: WHO passes resolution to eradicate smallpox globally.
 - 1967: Intensified eradication campaign launched.
 - 1977: The last natural case occurred in Somalia.
 - 1980: Smallpox certified eradicated worldwide.
- Why smallpox was suited for eradication:
 - Infected only humans (no animal reservoirs).
 - Symptoms were clearly distinguishable.
 - The vaccine was cheap, effective, and stable.

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



How it works:

Green: Vaccinated individuals forming a barrier.

Red: Vulnerable individuals (newborns, immunocompromised).

Gray: Pathogen source.

When enough people are vaccinated, pathogens cannot spread.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

- Vaccine impact is still incomplete for some diseases:
 - Pneumococcal Disease: 40.5% case reduction, 31.3% death reduction.
 - Hepatitis A: 89% case reduction, 88.7% death reduction.
 - Acute Hepatitis B: 83.9% case reduction, 83.6% death reduction.
- Barriers to complete control:
 - Global access gaps: Not everyone has access to life-saving vaccines.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- Summary of Impact and Future Priorities:
 - 100% Elimination Achieved: Diphtheria, smallpox, and both forms of polio have been eradicated in the US.
 - Timelines Collapsed: Vaccine development dropped from >100 years to <1 year, driven by mRNA and genomic tools.
 - Herd Immunity Works: Broad vaccination coverage effectively protects entire communities.
 - Persisting Coverage Gaps: Diseases like pneumococcal disease and hepatitis A/B highlight